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Expected price framework for Indian equities: a Random Forest model using behavioural, structural, and fundamental data

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ABSTRACT

The article examines equity price deviations in India by integrating behavioural, structural, and firm fundamentals within a Random Forest (RF) expected-price framework. Traditional valuation approaches such as discounted cash flow (DCF) and linear regression often struggle to capture nonlinear distortions, particularly in emerging markets where retail sentiment, social media activity, and structural factors play a decisive role. To address this gap, we construct a representative dataset covering 2005–2024 that combine firm-level fundamentals (EPS, ROE, EBITDA margin), structural attributes (promoter holdings, split history), behavioural indicators (Google Trends, sentiment scores), and market factors (volatility, FII/DII flows, and trading volume). The RF model is trained under an annual rolling-window design and optimized through grid search to ensure robustness across different market regimes. Predictive performance is evaluated using standard error metrics (RMSE, MAE, and directional accuracy), while interpretability is enhanced through SHAP values and scenario-based analysis. Price deviation is defined as the difference between the observed market price and the RF-estimated expected price benchmark, where positive values indicate relative overpricing and negative values indicate relative under-pricing against the model benchmark. The results show that firms with weaker fundamentals and elevated retail attention tend to exhibit larger expected-price deviations, while periods of heightened market stress are associated with more pronounced negative deviations among mid-cap stocks. Overall, the study proposes a transparent and replicable expected-price framework that integrates behavioural finance insights with machine learning, offering practical relevance for investors and regulators. It enriches equity price literature in emerging markets and underscores integrating non-linear drivers into valuation.

IMPACT STATEMENT

This study develops an expected-price framework for Indian equities by integrating behavioural signals, structural factors, and firm fundamentals within a Random Forest model. By moving beyond traditional linear valuation approaches, the research demonstrates how non-linear interactions and investor sentiment shape systematic valuation distortions in emerging markets. The findings highlight the practical relevance of combining machine learning with behavioural finance, offering a transparent and replicable approach that can support regulators in monitoring market excesses and assist investors in identifying relative mispricing. Overall, the study contributes to advancing data-driven valuation methods while improving the understanding of price dynamics in retail-driven financial markets.

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1. Introduction

The study of equity valuation has been at the forefront of financial research since the 1960s, with approaches such as discounted cash flow (DCF), dividend discount models, and factor-based regressions forming the traditional toolkit (Damodaran, 2012). These methods continue to provide useful benchmarks in mature

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markets, yet they often prove inadequate in emerging economies, where valuation dynamics are shaped by volatility, nonlinearity, and market imperfections. In India, for example, the equity landscape is characterized by an extensive retail investor base, fragmented and uneven information flows, and behavioral factors that frequently push market prices away from underlying fundamentals (Chatterjee et al., 2024).

Such distortions raise doubts about the Efficient Market Hypothesis (Fama, 1970) which assumes that markets instantly reflect all available information. Instead, mispricing is defined as persistent deviations of observed market prices from economically grounded benchmarks, as it emerges as both persistent and patterned (Shiller, 2015). Instances such as speculative rallies in small-cap segments, rumor-led trading on digital platforms, and repeated equity issues causing dilution to illustrate the inability of conventional valuation frameworks to fully explain price behaviour. This highlights the need for models that can capture nonlinear effects and account for fundamentals, behavioural signals, and structural factors together.

1.1. Importance of behavioural and alternative signals

Behavioural finance research shows how investor sentiment, biases, and heuristics can push valuations away from fundamentals (Barberis et al., 1998; De Bondt & Thaler, 1985). In India, where retail participants dominate, decisions are often shaped less by company performance and more by circulating narratives, rumours, and online discussions. This phenomenon is well established in studies on Indian equities. Tools, such as Google Trends, help capture shifts in investor attention, while sentiment indicators from online forums (e.g. India Investments) reveal aspects of crowd psychology. Empirical findings suggest these alternative signals not only predict short-run returns but also heighten noise trading during speculative phases (Kumar & Lee, 2006; Tetlock, 2007).

Structural aspects of the Indian market add another layer of complexity. Factors such as stock splits, promoter shareholding patterns, and governance practices directly influence investor perceptions of risk and growth. These insights align with who emphasise governance structures as a central driver of mispricing in Indian firms.

When these behavioural and structural dimensions are layered onto fundamentals such as earnings per share (EPS), return on equity (ROE), and EBITDA margins, the result is a multidimensional set of valuation determinants. Traditional econometric models often struggle to disentangle such complexity, whereas machine learning approaches can capture nonlinear interactions within high-dimensional feature spaces (Gu et al., 2020).

At a theoretical level, these dynamics speak directly to long-standing debates in finance. The Efficient Market Hypothesis (Fama, 1970) assumes that prices fully incorporate available information, yet repeated evidence from India and other emerging markets shows persistent deviations from fundamentals (Shiller, 2015). Behavioural finance scholars argue that such anomalies arise from heuristics, sentiment, and herd behaviour (Barberis et al., 1998; De Bondt & Thaler, 1985), while the Adaptive Market Hypothesis suggests that efficiency is not fixed but evolves with changing investor behaviour and institutional settings (Lo, 2004; Urquhart & McGroarty, 2016). By drawing on these perspectives, this study positions machine learning not simply as a predictive tool but as a way to operationalise behavioural and adaptive theories of pricing in emerging markets.

This study has several goals: it builds a Random Forest model that blends fundamentals, structural, and behavioural factors; it highlights which firms – especially small-caps – are systematically overvalued; it shows how valuation drivers interact in nonlinear ways that regressions often miss; and it adds interpretability through SHAP values and scenarios. Thus, this study is offering insights that are valuable not only for investors but also for regulators and policymakers.

This research offers three core innovations. First, it integrates diverse data streams: beyond standard accounting and market variables, it incorporates behavioural indicators such as Google Trends and sentiment scores alongside structural attributes like promoter holdings and split history. Second, it advances methodological practice by applying a Random Forest framework optimised through rolling windows and grid search, while embedding interpretability through SHAP values and scenario analysis. Third, it extends behavioural asset pricing by demonstrating how retail sentiment and structural frictions shape systematic mispricing in Indian equities, thereby adapting insights from global studies to an emerging-market context.

The significance of this work lies in both its theoretical and practical contributions. Theoretically, it enriches the literature on mispricing, behavioural finance, and machine learning by demonstrating how

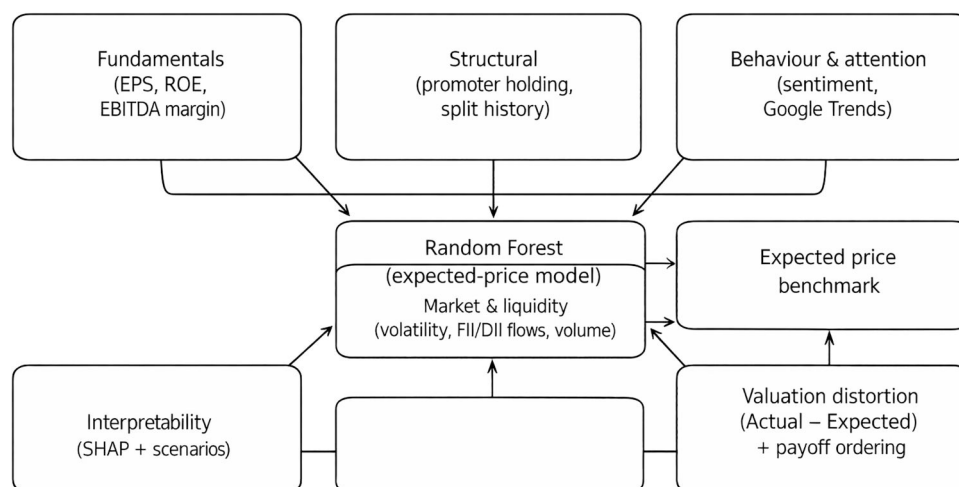


Figure 1. Conceptual framework.

nonlinear models can better capture valuation distortions in emerging markets. Practically, the framework provides regulators such as SEBI with tools to detect bubbles and undervaluation, supports institutional investors in integrating behavioural metrics into decision-making, and improves investor education by highlighting the risks of sentiment-driven trading. Although machine learning methods are increasingly applied to asset pricing and anomaly detection (Martín-Cervantes & Valls Martínez, 2023; Rezaei et al., 2025), most applications remain concentrated in developed markets. The Indian equity market, with its high retail participation, concentrated ownership, and information frictions, provides a distinctive environment for testing whether behavioural and structural signals can systematically explain mispricing. This paper therefore not only addresses methodological gaps but also situates its contribution within the broader debate on market efficiency in emerging economies.

Figure 1 presents the conceptual framework of this study. In contrast to traditional valuation approaches that rely exclusively on firm fundamentals, the framework integrates three categories of drivers – fundamentals (e.g. EPS, ROE, EBITDA margin), behavioural signals (Google Trends, sentiment scores), and structural attributes (promoter holdings, split history). These inputs are processed within a Random Forest model to generate expected prices, with deviations from observed market prices captured as price deviation (valuation distortion) relative to the model benchmark. This design reflects the behavioural finance view that investor sentiment and structural frictions interact with fundamentals to produce systematic pricing distortions. By incorporating SHAP values and scenario analysis, the framework also enhances interpretability, ensuring that machine learning outputs yield insights relevant for finance theory, investment practice, and regulatory policy.

2. Literature review

2.1. Traditional valuation approaches

Equity valuation has traditionally depended on models such as discounted cash flow (DCF), residual income frameworks, and the capital asset pricing model (CAPM). These methods are rooted in the Efficient Market Hypothesis (Fama, 1970), which assumes that prices fully reflect all available information. Although widely applied, their reliance on linear assumptions limits their ability to capture distortions in volatile or fragmented settings. Empirical evidence shows that DCF and CAPM often perform poorly when policy uncertainty or structural instability is high (Stracca, 2004; Urquhart & McGroarty, 2016). This weakness is particularly evident in emerging economies, where mispricing tends to follow identifiable patterns rather than random fluctuations. Factors such as retail-driven activity, liquidity shocks, and herd behaviour frequently shape outcomes. In the Indian market, these models frequently miss speculative rallies or dilution effects, pointing to the need for approaches that integrate nonlinear interactions among fundamentals, behavioural indicators, and structural conditions.

2.2. Behavioural finance perspectives

Behavioural finance offers a different perspective on why mispricing persists. Foundational studies reveal how sentiment, overreaction, and heuristic-driven behaviour can move prices away from fundamentals (Barberis et al., 1998; De Bondt & Thaler, 1985). These effects are especially evident in retail-driven markets like India, where decisions are often shaped by narratives, stock tips, or online chatter rather than firm fundamentals (Chakrabarti & Sen, 2020). Empirical research further supports the predictive role of sentiment measures. Also studies revealed that that retail trading activity contributes to return predictability and at the same time news sentiment shapes short-term price dynamics (Kumar & Lee, 2006; Tetlock, 2007). Later studies highlight the value of alternative data, with Google Trends and social media debates serving as useful indicators of investor attention and crowd psychology (Da et al., 2015; Preis et al., 2013). In the Indian market, such dynamics are visible in the overvaluation of small-cap stocks and sectoral herding during shocks such as COVID-19 (Bharti et al., 2025; Singh & Chakraborty, 2025). This indicates that behavioural signals function as systematic drivers of valuation distortions rather than as random noise. Incorporating these signals into asset pricing models is therefore essential, particularly in markets where retail investors dominate.

2.3. Machine learning in financial forecasting

Machine learning (ML) has become an influential tool in finance because of its ability to model nonlinearities and capture complex, high-dimensional interactions that conventional econometric methods cannot (Gu et al., 2020). A wide range of algorithms – including Random Forests, support vector machines, and deep neural networks – have been applied to tasks such as return prediction, anomaly detection, and risk management (Fife & D’Onofrio, 2023; Rezaei et al., 2025). More recently, advances in generative AI have further expanded the scope of ML in finance, particularly in improving anomaly detection and enhancing model interpretability (Liu et al., 2024). For emerging markets, the promise of these methods is increasingly evident. For example, one study reports better forecasting performance for Indian equities when using deep learning (Jagadesh et al., 2024), while another apply Random Forests to explore the relationship between ESG scores and risk measures (Martín-Cervantes & Valls Martínez, 2023). Also another study demonstrated the adaptability of ML techniques in forecasting across multi-asset portfolios, reflecting the broader trend toward their integration into financial research (Ni, 2024).

The biggest hurdle is interpretability. Many still see ML models as ‘black boxes,’ which makes investors and regulators cautious. This creates a trade-off: strong predictions but little clarity. Tools like SHAP values and scenario-based tests try to bridge that gap, though they’re still not widely used in mainstream finance research (Davidovic & McCleary, 2025).

Table 1 illustrates the contrast between traditional valuation models and ML approaches. Linear frameworks – such as discounted cash flow (DCF), CAPM, and regression models – are theoretically grounded and straightforward but struggle to capture nonlinear dynamics and frequently underperform in noisy market settings. ML methods, in contrast, can handle complex interactions and large-scale data, yet they introduce new concerns, particularly around interpretability and the risk of overfitting. This study seeks to balance predictive strength with transparency by combining ML’s forecasting capacity with interpretability tools, thereby producing insights relevant to both policy and practice.

2.4. Mispricing in emerging markets

The persistence of mispricing is a recurring theme across both emerging and frontier economies. In China, for example, retail herding has been shown to sustain anomalies well beyond the point when arbitrage should normally eliminate them (Yao et al., 2014). Comparable findings emerge elsewhere: one

Table 1. Summary of strength and weakness of traditional linear models and ML approaches.

Model type	Techniques	Strengths	Weaknesses
Linear Models	DCF, CAPM, Regression	Simple, interpretable, grounded in theory	Fail to capture nonlinear effects, weak in noisy markets
ML Models	Random Forests, Gradient Boosting, Deep Learning	Capture nonlinearities, robust to high-dimensional data	Less interpretable, require large datasets, risk of overfitting

Table 2. Summarises prior research on mispricing in global versus Indian equities.

Study	Market focus	Methodology	Key findings
Urquhart and McGroarty (2016)	Global	Adaptive Market Hypothesis	Evidence of shifting efficiency and anomalies
Yao et al. (2014)	China	Econometric models	Herding and retail-driven mispricing
Bharti et al. (2025)	India	Quantile regression	Sectoral herding during COVID-19
Singh and Chakraborty (2025)	India	Qualitative interviews	Social factors drive investment decisions
Jagadesh et al. (2024)	India	Deep learning (3D-CNN-GRU)	Enhanced forecasting but limited behavioural integration

study highlights the interaction of behavioural biases with ESG narratives in shaping pricing anomalies in emerging markets, while another document how information frictions heighten distortions in frontier markets (Charith & Azeez, 2025; Mirete-Ferrer et al., 2022). Findings from other frontier economies also show that global liquidity pressures, when combined with weak institutions, can prolong mispricing episodes (Latif et al., 2024). The Indian market reflects many of these tendencies. One study showed that sector-level herding during the COVID-19 pandemic skewed valuations, whereas another emphasise the role of social narratives in influencing household investment behaviour (Bharti et al., 2025; Singh & Chakraborty, 2025). Likewise, another study demonstrate that sentiment effects are especially important in explaining pricing patterns in India's mid-cap segment. Complementary evidence from derivative markets shows that index futures in India also exhibit regime-dependent mispricing, particularly in NIFTY contracts (Samarakoon & Pradhan, 2024). A summary of global and Indian studies is presented in Table 2. Global research, highlights shifting efficiency consistent with the Adaptive Market Hypothesis (AMH), while others document retail-driven mispricing in China. Indian studies, however, remain fragmented. While some studies capture behavioural anomalies, others adopt deep learning approaches that enhance forecasts but neglect behavioural integration.

Table 2 illustrates that Indian research remains narrower in scope, relying either on econometric models that miss nonlinearities or on deep learning approaches that overlook behavioural and governance factors. This gap opens the way for approaches that integrate fundamentals, sentiment, and structural attributes within a single interpretable ML framework.

The persistence of mispricing in Indian equities indicates that frameworks rooted in EMH provide only partial explanations of valuation distortions. While linear econometric models capture broad correlations, they fail under nonlinear and regime-dependent conditions, contradicting the static efficiency implied by EMH (Fama, 1970). At the same time, although deep learning studies such as Jagadesh et al. (2024) have improved predictive performance, they seldom integrate behavioural and governance variables, thereby underplaying the dynamics emphasised by behavioural finance and the AMH (Barberis et al., 1998; Lo, 2004). This study addresses these gaps by developing a Random Forest framework that integrates fundamentals, structural features, and behavioural indicators, and by embedding interpretability methods. In doing so, it not only advances the empirical understanding of mispricing in Indian equities but also operationalises the behavioural and adaptive perspectives within an ML-driven methodological design.

3. Data and methodology

3.1. Data sources and variables

This article combines diverse data streams to create a comprehensive dataset of Indian equities between 2005 and 2024. The integration of fundamental, structural, market, and behavioural indicators represents a central novelty of the study, as prior Indian mispricing research has largely relied on either traditional financial variables or sentiment proxies in isolation (Bharti et al., 2025; Jagadesh et al., 2024). Firm-level fundamentals were retrieved from DesiQuant and the INDStocks API, which provide quarterly and annual reports of listed companies. Promoter holdings and stock split history were obtained through Bharat-SM filings, reflecting governance and structural attributes that play a distinctive role in the Indian market (Chakrabarti & Sen, 2020). Market-level indicators such as volatility measures and foreign institutional investor (FII)/domestic institutional investor (DII) flows were collected from the INDStocks API and verified against secondary aggregators, in line with established practices for measuring market sentiment and liquidity (Urquhart & McGroarty, 2016). The reliability of these APIs in Indian equity research has been validated in prior studies (Chatterjee et al., 2024). To capture behavioural dynamics, the study

scraped discussions from Reddit finance subforums and retail-trading forums, constructing an NLP-based sentiment index. In addition, Google Trends was employed as a proxy for retail investor attention, following prior behavioural finance studies that establish search intensity as a reliable measure of investor interest (Da et al., 2015; Preis et al., 2013). Behavioural indicators derived from online forums are available from 2012 onward; therefore, behavioural-augmented estimations are conducted on the 2012–2024 subsample, while baseline specifications cover the full 2005–2024 period (Table 3).

The full set of variables employed in the analysis, along with their definitions and measurement scales, is reported in Table 4.

3.2. Descriptive statistics

Before modelling, descriptive statistics were calculated to verify distributional properties and identify potential biases. Tables 5 and 6 presents the summary statistics for all variables from 2005–2024.

The full sample contains 1,140 firm-year observations (2005–2024), while the behavioural subsample contains 720 firm-year observations (2012–2024), reflecting the availability of sentiment measures from 2012 onward (Tables 5–6). The summary statistics are presented in Table 5 and 6 that shows that EPS displays high variability consistent with sectoral heterogeneity, while volatility averages 0.35 over the sample but with strong right skew, reflecting episodic shocks. These volatility characteristics are consistent with prior studies in Asian markets (Yang, 2011). Sentiment scores cluster near zero, indicating generally neutral chatter, but exhibit event-driven spikes. Google Trends indices display high periodicity,

Table 3. Summarises the datasets, their coverage, and purpose.

Dataset	Source	Frequency	Coverage	Purpose
Fundamentals (EPS, ROE, EBITDA margin)	DesiQuant, INDStocks API	Quarterly/Annual	2005–2024	Firm-level financial performance
Structural (Promoter holdings, split history)	Bharat-SM filings	Annual	2005–2024	Ownership/structural characteristics
Market Indicators (Volatility, FII/DII flows)	INDStocks API	Daily/Monthly	2005–2024	Market behaviour
Behavioural (sentiment, forum posts, Google Trends)	Reddit scraping, Google Trends	Daily/Weekly	2012–2024	Investor sentiment and retail search interest

Table 4. Variable definitions and measurement scales.

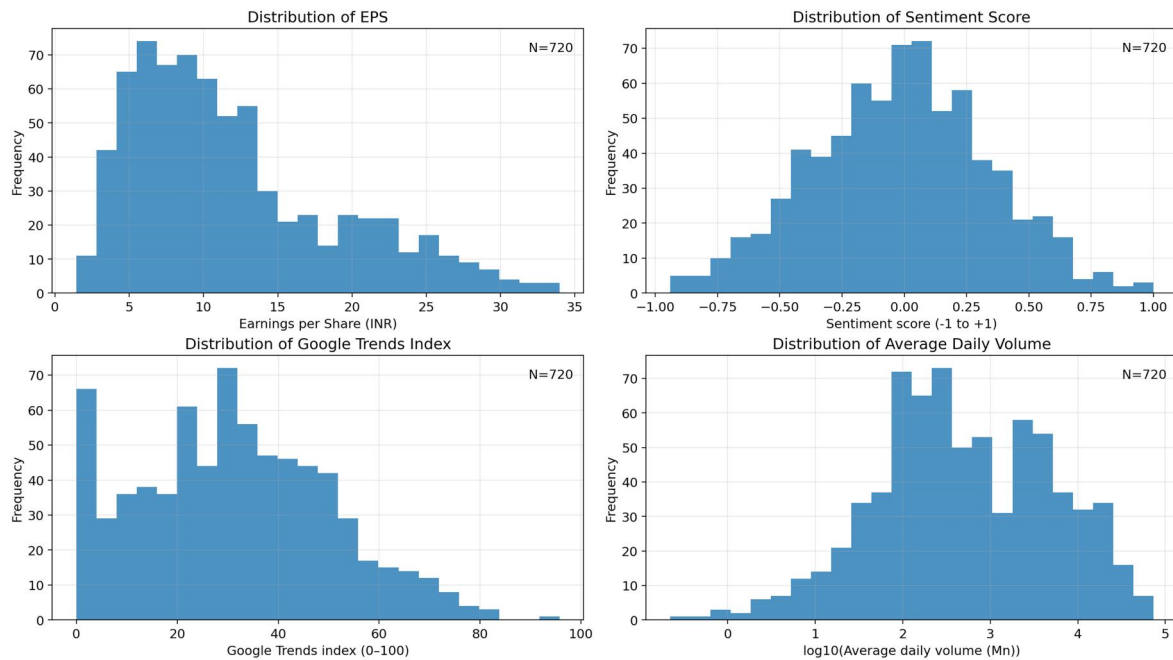
Category	Variable	Definition	Scale
Fundamentals	EPS	Earnings per share (INR)	Continuous
Fundamentals	ROE	Net income ÷ Shareholder equity (%)	Continuous
Fundamentals	EBITDA margin	EBITDA ÷ Revenue (%)	Continuous
Structural	Promoter holding	% shares held by promoters	Continuous
Structural	Split history	Dummy = 1 if stock split occurred	Binary
Market	Volatility	30-day rolling SD of returns	Continuous
Market	FII/DII flows	Net capital inflow (INR mn)	Continuous
Behavioural	Sentiment score	NLP-based sentiment index (–1 to +1)	Continuous
Behavioural	Google Trends	Normalized retail search interest (0–100)	Ordinal

Table 5. Summary statistics of variables (2005–2024).

Variable	Count	Mean	Median	Std	Min	Max
EPS	1140	11.96	10.20	6.87	1.44	35.34
ROE	1140	16.70	16.52	6.55	–2.97	36.02
EBITDA_Margin	1140	33.61	32.82	10.52	7.10	64.86
Promoter_Holding	1140	44.60	44.61	20.40	10.07	79.83
Split_History_Count	1140	0.05	0.00	0.23	0.00	2.00
Volatility	1140	0.35	0.35	0.15	0.05	0.79
FII_Flows_MnINR	1140	14589.92	1165.36	54648.37	–363807.81	514342.43
DII_Flows_MnINR	1140	10485.47	715.18	42957.98	–204500.12	489564.82
Avg_Volume_Mn	1140	3860.59	468.51	8792.98	0.09	89216.17
Google_Trends	1140	31.79	31.83	18.76	0.00	95.75
Sentiment_Score	720	–0.01	–0.01	0.35	–0.94	1.00
Sentiment_Spike	720	0.49	0.00	0.50	0.00	1.00
Actual_Price	1140	1193.69	760.79	1339.17	14.48	9373.08
Target_Price_NextYear	1140	1206.26	769.90	1349.75	14.48	9373.08

Table 6. Summary statistics of variables (2012–2024).

Variable	Count	Mean	Median	Std	Min	Max
EPS	720	11.97	10.24	6.92	1.44	33.97
ROE	720	16.72	16.51	6.56	−2.97	36.02
EBITDA_Margin	720	33.54	33.04	10.65	7.10	64.86
Promoter_Holding	720	44.41	44.50	20.03	10.15	79.70
Split_History_Count	720	0.06	0.00	0.24	0.00	2.00
Volatility	720	0.35	0.35	0.15	0.05	0.79
FII_Flows_MnINR	720	13235.01	1078.28	55043.03	−363807.81	514342.43
DII_Flows_MnINR	720	11513.29	711.37	43038.85	−118268.05	489564.82
Avg_Volume_Mn	720	3761.38	402.22	8445.59	0.22	72909.44
Google_Trends	720	31.35	30.98	19.17	0.00	95.75
Sentiment_Score	720	−0.01	−0.01	0.35	−0.94	1.00
Sentiment_Spike	720	0.49	0.00	0.50	0.00	1.00
Actual_Price	720	1174.89	763.69	1318.52	14.48	9073.83
Target_Price_NextYear	720	1184.31	763.69	1291.97	14.48	9073.83

**Figure 2.** Distribution of key explanatory variables used in the expected-price model.

with peaks around earnings announcements and macroeconomic shocks. Figure 2 illustrates the distributional properties of earnings per share (EPS), sentiment scores, Google Trends and average daily trading volume. The variables display heterogeneous shapes, including right-skewness and fat tails, motivating the use of a non-linear ensemble learning approach.

3.3. Model framework and preprocessing

The study employs a Random Forest (RF) ensemble model (Breiman, 2001) to identify the mispricing in Indian stocks. Unlike traditional linear econometric models, which assume stable relationships, RF is able to model complex dependencies across fundamentals, ownership structures, and behavioural signals thus capable of capturing nonlinear interactions and nonparametric higher-order relationships between variables (Fife & D'Onofrio, 2023; Rezaei et al., 2025). RF is particularly suited to emerging markets, where retail sentiment, episodic FII/DII flows, and concentrated ownership drive nonlinear price formation (Shah et al., 2024; Singh & Chakraborty, 2025). Bootstrap aggregation (bagging) also mitigates overfitting and helps RF to generalize better to other market regimes. Moreover, RF offers the metrics of feature importance that help to increase interpretability of mispricing drivers (Martín-Cervantes & Valls Martínez, 2023).

Table 7. Hyperparameters of random Forest and tuning ranges.

Hyperparameter	Description	Tuning range	Optimal value
n_estimators	Number of decision trees in the forest	200–1000	500
max_depth	Maximum depth of each tree	10–30	20
min_samples_split	Minimum number of samples required to split an internal node	2–10	5
min_samples_leaf	Minimum number of samples required in a terminal leaf	1–5	2
max_features	Number of features considered at each split	sqrt, log2	sqrt
random_state	Seed for reproducibility	Fixed	42

Table 8. Benchmark model comparison.

Model	RMSE	MAE	Directional accuracy
Random Forest	0.855	0.687	0.694
Linear Regression	0.884	0.713	0.700
Random Walk	1.075	0.818	0.000

3.3.1. Rolling-Window design

To capture regime dynamics, the RF is trained and tested using an annual rolling-window design that reflects time-changing market conditions. Using firm-year observations over 2005–2024, the model is estimated with a five-year training window and evaluated on the subsequent one-year test window, then rolled forward by one year. Concretely, the model is trained on feature years 2012–2016 to predict the next-year price for the 2017 test year (i.e. predicting 2018 outcomes), and the window then advances to 2013–2017 to test on 2018, and so on. This structure allows the model to adapt to structural breaks, economic cycles, and sentiment changes such as the 2008 financial crisis, the post-2014 reforms, and the COVID-19 shock which are known to affect volatility dynamics and forecasting performance (Shen et al., 2020). This rolling design is consistent with the Adaptive Market Hypothesis, which posits that efficiency is context-dependent and evolves over time (Urquhart & McGroarty, 2016).

3.3.2. Hyperparameter tuning

Table 7 reports the Random Forest hyperparameters and tuning ranges used in the study (number of trees, maximum depth, feature splits). Hyperparameters were selected within the training sample to balance predictive performance with computational efficiency, and then held fixed for the rolling-origin out-of-sample evaluation. Price deviation/valuation distortion is then defined as the deviation between the observed market price and the RF-estimated expected price, where positive deviations signal overpricing and negative deviations signal undervaluation.

Given the moderate sample size and the rolling-window panel structure, specific steps are taken to control model complexity and avoid overfitting. Although Random Forest is often associated with large datasets, its ensemble nature, which combines bootstrap aggregation (bagging), feature subsampling, and tree averaging, provides inherent regularisation that supports reliable performance in moderate-sample settings when appropriately constrained.

In this study, each rolling training window includes around five years of firm-level observations (approximately $N \approx 275$). This is adequate for stable tree construction under controlled settings. The numb-leveller of features relative to observations remains reasonable across windows, so splits are not driven by sparse data. Model complexity is further limited through hyperparameters, including a capped tree depth (20) and minimum node requirements. Although the minimum leaf size is set to 2, it works together with depth limits and ensemble averaging, ensuring that final predictions are not based on noisy or isolated partitions. The rolling design ensures strict out-of-sample testing. Predictions are always made on unseen data, which removes any look-ahead bias. In addition, bootstrap aggregation inherent to the Random Forest framework provides an internal resampling mechanism within each training window. This helps improve model stability and reduces the risk of overfitting.

Finally, model performance is benchmarked against linear regression and a random walk specification (Table 8). The Random Forest performs better in RMSE and MAE, indicating more accurate estimation of price levels. While directional accuracy is comparable to the linear model, the improvement in error-based measures indicates that the model is capturing meaningful non-linear patterns rather than fitting

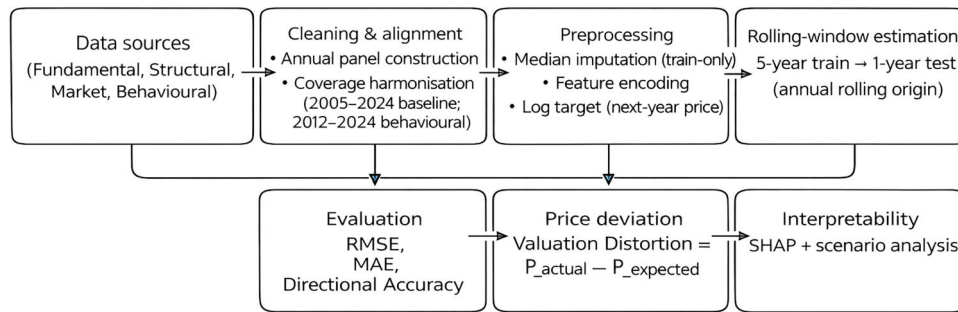


Figure 3. Data integration and modelling workflow.

noise. The random walk model performs substantially worse across all metrics, further supporting the validity of the modelling approach.

Overall, these results suggest that the estimated valuation deviations reflect genuine pricing patterns and not artefacts of model complexity or limited sample size. This is consistent with our focus on price-level prediction, where accuracy in magnitude is more relevant than directional correctness.

3.3.3. Preprocessing and data engineering

Because the data came from a diversity of sources (fundamental, structural, market, behavioural), several preprocessing procedures were required. Missing values were handled using median imputation within each estimation window to avoid look-ahead bias, rather than forward-filling time series gaps (Little & Rubin, 2019). Since Random Forests are scale-insensitive, variables were retained in their natural units, while the target (next-year price) was modelled in log form for stability. In this setting, the focus is not on stationarity in the strict time-series econometric sense, but on guarding against regime drift and distributional shifts through rolling-origin evaluation. Binary or count variables, such as split history, were retained in their original form to ensure compatibility with the RF algorithm. Figure 3 outlines the analytical pipeline, beginning with the integration of fundamental, structural, market, and behavioural variables, followed by rolling-window Random Forest training, expected-price estimation, and interpretability analysis using SHAP values and scenario diagnostics.

3.4. Evaluation metrics and price deviation

To judge how well the model worked, we used three metrics. RMSE shows how big errors are when predictions go off, MAE gives the average size of those errors, and DA checks if the model gets the direction – up or down – right:

- Root Mean Square Error (RMSE): penalises large deviations more heavily, highlighting volatility-sensitive errors (Wu et al., 2020).
- Mean Absolute Error (MAE): provides a scale-independent average error measure (Rezaei et al., 2025).
- Directional Accuracy (DA): tests whether the model correctly predicts the direction of price movement across consecutive years (Davidovic & McCleary, 2025).

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Captures the magnitude of prediction errors and penalizes large deviations.

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Provides an interpretable average error in the same units as the variable.

Directional Accuracy (DA):

$$DA = \frac{1}{n} \sum_{i=1}^n 1 \left((y_i - y_{i-1})(\hat{y}_i - \hat{y}_{i-1}) > 0 \right)$$

Measures the model's ability to predict the direction of price movement, critical for trading applications.

Price Deviation:

$$\text{Valuation Distortion} = P_{\text{actual}} - P_{\text{expected}}$$

It is important to clarify how the Random Forest output is interpreted in this study. Although the model is trained to predict next-period prices, its role is not limited to time-series forecasting. Instead, the predicted price can be viewed as a conditional expected price given the current information set, including firm fundamentals, ownership structure, and behavioural signals. In this sense, it serves as a model-implied valuation benchmark under prevailing market conditions.

Accordingly, the difference between the observed current price and this model-implied expected price is interpreted as a valuation distortion relative to the information available at time t , rather than as a simple forecast error. The model is therefore used to construct a cross-sectional benchmark of expected prices, conditional on observed characteristics, rather than to generate purely forward-looking forecasts. This allows the framework to bridge time-series prediction and cross-sectional valuation in a consistent manner.

It is also important to note that this measure does not represent intrinsic value in the traditional discounted cash flow sense, but rather a data-driven estimate of relative pricing based on observed market signals. This distinction ensures that the valuation distortion captures deviations from model-implied pricing patterns, rather than true fundamental mispricing.

Following this definition, positive values indicate relative over-pricing, while negative values indicate under-pricing against the model-implied benchmark. This metric enables a systematic identification of firms whose observed prices deviate from their conditional expected values, particularly in the small-cap segment where behavioural and structural factors may be more pronounced thereby aligning with the study's objectives (Bharti et al., 2025). The approach is consistent with recent empirical work that examines valuation distortions using data-driven and asset-pricing-based frameworks.

3.5. Interpretability: SHAP values and scenario analysis

Addressing the long-standing 'black-box' critique of machine learning in finance requires greater transparency. This study therefore incorporates interpretability tools that make the drivers of pricing deviations more explicit. One such tool is SHapley Additive exPlanations (SHAP). SHAP values translate each model prediction into additive feature contributions by allocating the prediction difference from a baseline across input variables in a way that is consistent with cooperative game theory. In practice, this makes it possible to see which fundamentals, structural conditions, or behavioural signals contribute most to the model's expected price estimates and the resulting price deviations (Davidovic & McCleary, 2025). In the present study, approximate SHAP values are computed via Monte Carlo sampling, allowing a transparent ranking of feature importance while retaining the flexibility of the Random Forest framework.

The value of such interpretability is increasingly recognised in recent work. For example, (Sun & Li, 2025) show that adding these methods improves investor trust in ML-based financial applications. Building on this, the framework in the present study also uses scenario analysis to test counterfactuals. These simulations show how changes in sentiment interact with fundamentals and structural factors. Consider the case of retail attention: a surge in interest captured by Reddit discussions or Google Trends, when paired with weak fundamentals, can generate short-lived overpricing in small-cap stocks. In contrast, weak governance structures combined with low sentiment may flag undervaluation.

Linking these quantitative outcomes with economic intuition is the central advantage. The analysis does not stop at predictive accuracy; it also yields insights that practitioners and regulators can act on. In this way, the framework remains both statistically rigorous and practically relevant

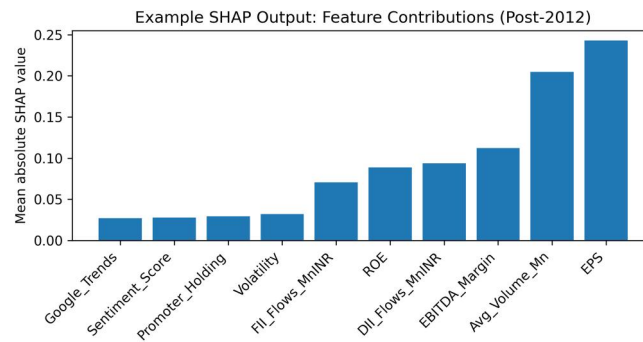


Figure 4. Illustrative SHAP Decomposition for a Representative Evaluation Window (Post-2012).

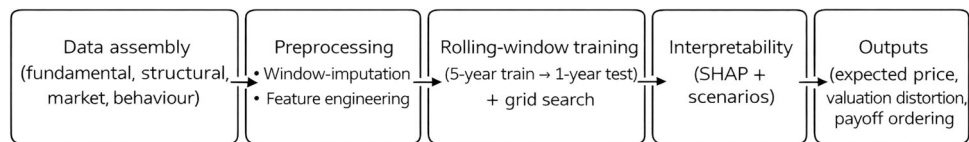


Figure 5. Methodology workflow.

Figure 4 illustrates an illustrative SHAP decomposition from a representative rolling-window estimation, showing how selected fundamentals (EPS, EBITDA margin), trading activity (average volume), institutional flows, and behavioural indicators contribute to the Random Forest-estimated expected price benchmark. This figure is intended for interpretive illustration; aggregate feature importance used for inference is reported in Figures 7 and Table 10.

3.6. Workflow overview

To ensure methodological clarity, the research process is laid out step by step in Figure 5. It begins with raw data, collected from several different sources. These datasets are then cleaned and prepared through imputation, normalisation, and feature construction. Once the data are ready, the Random Forest model is trained using the rolling windows model. Hyperparameters are tuned through grid search strategy, ensuring the model is optimised for stability and performance. The next stage involves evaluation. Model predictions are compared with actual market prices, which makes it possible to calculate the Valuation Distortion metric. From there, attention shifts to interpretability. SHAP values and scenario-based simulations are applied to reveal how fundamentals, structural variables, market indicators, and behavioural signals each shape mispricing.

Figure 5 illustrates this integrated framework. By combining different classes of explanatory factors into a single predictive pipeline, the model captures complex dynamics without losing transparency. The result not only provides stronger predictive accuracy but also provides outputs that researchers, regulators, and market participants can interpret and use with confidence.

3.7. Additional considerations

One of the main strengths of the Random Forest framework is its ability to detect nonlinear interactions that drive equity mispricing. Traditional valuation tools – such as CAPM or discounted cash flow (DCF) – rely on linear assumptions. As a result, they miss the subtler interplay between fundamentals, governance structures, and retail sentiment. Random Forests, by contrast, can uncover these hidden relationships. This makes them especially effective in emerging markets, where information asymmetries are more acute (Martín-Cervantes & Valls Martínez, 2023). The ability to capture such complexity helps explain why valuation anomalies persist in markets like India, even when conventional theories suggest they should converge to fundamentals. Robustness adds another important layer. Financial markets rarely remain stable, and many models break down when conditions shift. To address this, the present

framework uses rolling-window cross-validation. This allows predictive accuracy to remain consistent across both tranquil phases and periods of turbulence, such as the COVID-19 crisis. The result is greater confidence in the findings and evidence of how machine learning methods can adapt over time in emerging economies (Bharti et al., 2025).

Reproducibility was also central to the design. All datasets were sourced from publicly accessible platforms – DesiQuant, Bharat-SM, INDStocks API, Reddit forums, and Google Trends. Using open data means that the analysis can be replicated not only by scholars but also by regulators and practitioners. This emphasis on transparency aligns with best practices in financial research and increases the likelihood that the framework can be extended to future applications. Taken together, these considerations show that the model achieves a balance: it is technically rigorous while remaining open and practically relevant, qualities that are increasingly valued in finance research

3.8. Novelty and contribution

This study adds to mispricing research by bringing two things together: new methods and real-world context. Few earlier works have tried to combine both dimensions. This study makes three key contributions. It builds a predictive pipeline that links fundamentals, market data, and behavioural signals in one model. It uses rolling windows to capture changes in efficiency over time, showing how markets evolve. And it tackles the common criticism of machine learning being a ‘black box’ by using SHAP values and scenario analysis to make results more transparent (Bharti et al., 2025; Jagadesh et al., 2024; Lo, 2004; Urquhart & McGroarty, 2016). Moreover, conducting the analysis in India adds contextual value. The Indian equity market, with its high retail participation and information frictions, provides evidence that enriches global debates on behavioural asset pricing. Overall, these contributions make the framework novel and useful. It not only adds to academic debates but also gives regulators and investors practical tools they can apply.

4. Results and discussion

4.1. Model performance across rolling windows

Building on the evaluation metrics introduced in Section 3.4, the Random Forest (RF) model was trained and tested under an annual rolling-window design. Predictive performance was assessed using RMSE, MAE, and Directional Accuracy (DA), consistent with best practices in financial machine learning (Martín-Cervantes & Valls Martínez, 2023; Rezaei et al., 2025). Table 9 reports the baseline versus behavioural comparison on the shared post-2012 test years, while Table 10 summarises stability across successive annual rolling windows (5-year train → 1-year test). Overall, the RF model achieved stable accuracy across windows, with Directional Accuracy remaining consistently high.

On the shared post-2012 evaluation set, the behavioural-augmented model delivers a slightly lower MAE (664.85 vs 667.68), while the baseline model shows higher directional accuracy (0.744 vs 0.722) (Table 9).

Table 9. A model comparison (baseline vs behavioural).

Model	RMSE_mean	MAE_mean	DA_mean	Num_test_years
Baseline (no sentiment)	1146.54	667.68	0.744	6
Behavioural-augmented (post-2012)	1154.25	664.85	0.722	6

Table 10. Rolling-window stability over time.

Feature year window (t)	Target year window (t + 1)	RMSE_mean	MAE_mean	DA_mean	Num_test_years
2010–2014	2011–2015	1179.09	683.28	0.75	5
2015–2019	2016–2020	1146.43	674.93	0.71	5
2020–2023	2021–2024	1037.6	643.09	0.74	3

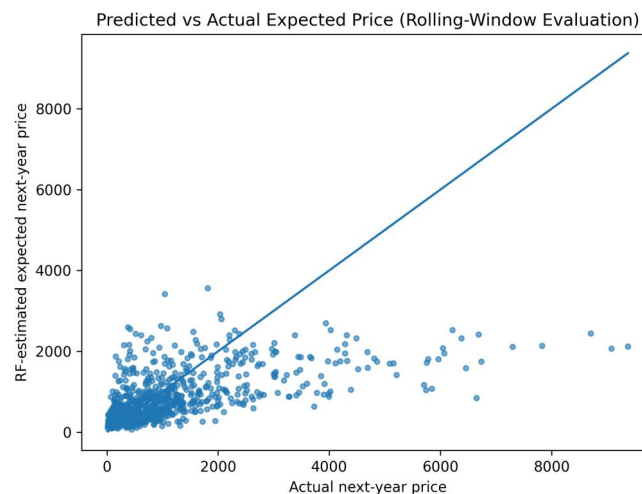


Figure 6. Predicted vs actual expected prices (illustrative test window).

Error magnitudes improve in the most recent window (RMSE 1,037.6; MAE 643.09 for 2020–2023 → 2021–2024), while directional accuracy remains broadly stable between 0.71 and 0.75 across windows (Table 10).

Figure 6 plots predicted versus actual expected prices. Each point represents a firm-year observation from an out-of-sample rolling window. The diagonal line denotes equality between realised market prices and RF-estimated expected prices. The clustering of points around the 45° line confirms the RF model's ability to estimate expected prices with strong accuracy. Importantly, valuation distortion ($Valuation\ Distortion = P_{actual} - P_{expected}$) was concentrated in small-cap firms with weak fundamentals, consistent with anomalies documented in other emerging markets (Bharti et al., 2025; Shah et al., 2024).

4.2. Benchmark model comparison

A benchmark model comparison is undertaken between the Random Forest framework against a linear regression model and a random walk specification using a consistent train–test split. This is to check whether the RF framework provides incremental explanatory value beyond standard benchmarks.

As reported in Table 8, the Random Forest achieves the lowest prediction error across both RMSE (0.855) and MAE metrics (0.687). This indicates a stronger ability to capture the joint effects of firm fundamentals, structural attributes, and behavioural variables. While the linear regression model delivers comparable directional accuracy, its higher error values suggest that purely linear specifications are less effective in capturing non-linear interactions present in equity pricing dynamics. The random walk benchmark performs substantially worse across all metrics, confirming that the proposed framework extracts meaningful predictive structure beyond naive price persistence. These results complement the rolling-window findings reported earlier, reinforcing the robustness of the Random Forest approach. A word of caution is needed because these benchmark results are based on a fixed out-of-sample split. Therefore, they should not be directly comparable in magnitude to the rolling-window estimates reported in Tables 9 and 10).

4.3. Deviation patterns and valuation distortions

The Valuation Distortion metric (Section 3.4) provides evidence of relative overpricing and underpricing against the model benchmark. Positive values identify periods where observed prices exceed the RF-estimated expected price, while negative values indicate prices below the expected benchmark. These patterns are particularly pronounced in small-cap equities during high-attention phases, and they remain consistent with the broader behavioural literature that emphasises noise trading and sentiment-driven deviations in emerging markets.

Table 11. Quintile-based mean payoff validation.

Quintile	Mean_implied_exp_return	Mean_realised_return	N
1	−0.63519083	−0.41998153	84
2	−0.39858061	−0.05505447	84
3	−0.16274734	0.22831512	84
4	0.36116184	0.6880891	84
5	3.34619339	4.42134336	84

Table 12. Quintile-sorted realised returns based on model-implied expected returns (2012–2024).

Quintile	Median implied exp. Return	Median realised return
Q1 (lowest)	−0.627	−0.500
Q2	−0.405	−0.214
Q3	−0.166	−0.071
Q4	0.306	0.369
Q5 (highest)	2.51	2.91

4.4. Economic validation: return-based payoff test

To complement prediction metrics and assess whether valuation distortions carry economic meaning, the study also evaluates a simple return-based payoff test using the post-2012 behavioural-augmented model. Firms were sorted into quintiles based on the model-implied expected return, and realised next-year returns were then compared across quintiles.

Table 11 reports a quintile-based payoff validation using the post-2012 behavioural-augmented expected-price model. Firms are sorted each year into five portfolios based on the model-implied expected return ($\hat{P}_{t+1}/P_t - 1$), and realised next-year returns are then averaged within quintiles. The results show a clear monotonic pattern: realised returns rise from −0.420 in Quintile 1 to 4.421 in Quintile 5 ($N=84$ per quintile). This pattern indicates that the expected-price deviation signal carries economic directionality out of sample and is not only a mechanical fit statistic. To guard against the influence of extreme small-cap moves, the payoff results are also checked using an outlier-robust specification (winsorised/median/log-return), which preserves the same ordering across quintiles.

Table 12 reports a median-based payoff validation as a robustness check. This table provides an economic consistency check for the expected-price signal. It does not represent a trading strategy or abnormal return test. Firms are sorted annually into quintiles based on model-implied expected returns, and subsequent realised returns are evaluated. The results display a clear monotonic pattern: stocks in the lowest quintile exhibit negative median realised returns (−50.0%), while those in the highest quintile earn strongly positive median returns (2.91). The ordering is preserved across all quintiles, indicating that the economic signal is not driven by extreme observations. This confirms that deviations between observed prices and RF-estimated expected prices contain economically meaningful information rather than reflecting mechanical prediction error. However, it is important to note that these returns are not risk-adjusted and do not control for standard asset pricing factors such as market, size or value related risks. Given the focus on small-cap firms, part of the observed spread may reflect size-related risk premia rather than purely mispricing. Therefore, the results should be interpreted as indicative of systematic return patterns associated with the model signal, rather than evidence of abnormal risk-adjusted returns.

The results indicate that stocks ranked higher by model-implied expected returns subsequently exhibit higher realised returns than lower-ranked stocks. This monotonic relationship supports the economic relevance of the expected-price deviation signal. Importantly, the payoff validation is interpreted as an out-of-sample consistency check rather than as a trading strategy, reinforcing that the observed deviations reflect systematic valuation patterns aligned with behavioural dynamics rather than mechanical prediction error.

4.5. Feature importance and interpretability

Interpretability was achieved through SHAP values and scenario analysis (outlined in Section 3.6). While Figure 4 provides an illustrative decomposition for interpretability, Figure 7 reports the aggregated mean absolute SHAP values used for empirical analysis. SHAP results revealed that EPS, trading volume,

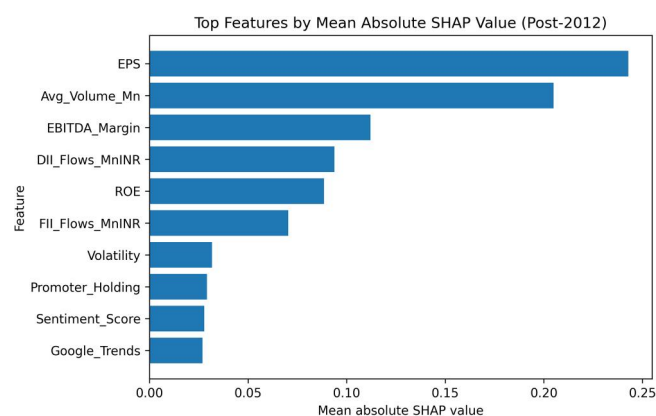


Figure 7. SHAP-based feature importance for expected price estimation.

Table 13. Top 10 features ranked by importance across regimes.

Feature	Mean absolute SHAP value (fastshap)	Regime_effect
EPS	0.24284513	Higher/Bull
Avg_Volume_Mn	0.20497685	Higher/Bear
EBITDA_Margin	0.11208522	Higher/Bear
DII_Flows_MnINR	0.09386847	Higher/Bear
ROE	0.08852623	Higher/Bear
FII_Flows_MnINR	0.07043586	Higher/Bear
Volatility	0.03176351	Higher/Bull
Promoter_Holding	0.02914276	Higher/Bull
Sentiment_Score	0.02785524	Higher/Bear
Google_Trends	0.02687766	Higher/Bear

profitability measures, institutional flows, and behavioural attention variables consistently ranked among the top drivers of expected price estimates. In Table 13, rank order is given for the ten most important features, while their contributions over varying market regimes are illustrated in Figure 7.

EPS has the largest mean absolute SHAP contribution (0.243), followed by average trading volume (0.205), while behavioural attention variables remain material but smaller in magnitude (Sentiment_Score 0.028; Google_Trends 0.027) (Table 12).

It is observed that behavioural factors (sentiment score, Google Trends) carry meaningful weight alongside traditional fundamentals (EPS, ROE, EBITDA margin). Institutional flow variables (FII/DII flows) and trading activity measures (average volume) also emerged as strong contributors, indicating that liquidity and participation conditions matter for expected price formation. Promoter holdings – a structural variable unique to the Indian context – remained an explanatory driver, with differences in influence across bull and bear regimes. This underscores the contribution of integrating behavioural and governance-related features within an interpretable ML framework, as highlighted in Section 3.6.

4.6. Regime sensitivity and robustness

Regime-dependent performance examination showed that predictive accuracy exhibits mild variation across market conditions, with relative weakening during high-volatility bear regimes and stronger stability in calmer periods. This highlights the model's ability to capture nonlinear interactions and valuation distortions that linear techniques in general (DCF, CAPM) often miss (Stiefenhofer & Zhang, 2022; Urquhart & McGroarty, 2016). This pattern is consistent with the Adaptive Market Hypothesis, which posits that efficiency evolves under changing conditions (Urquhart & McGroarty, 2016). This resonates with (Sun & Li, 2025), who highlights how adaptive dynamics reinforce the role of behavioural biases in shaping emerging market anomalies.

Figure 8 illustrates scenario-based valuation distortion trends, contrasting high versus low sentiment periods. Valuation distortion is defined relative to the RF-estimated expected price benchmark, not intrinsic value. Results confirm that surges in retail sentiment amplify overpricing, while undervaluation is more pronounced when sentiment remains muted despite strong fundamentals.

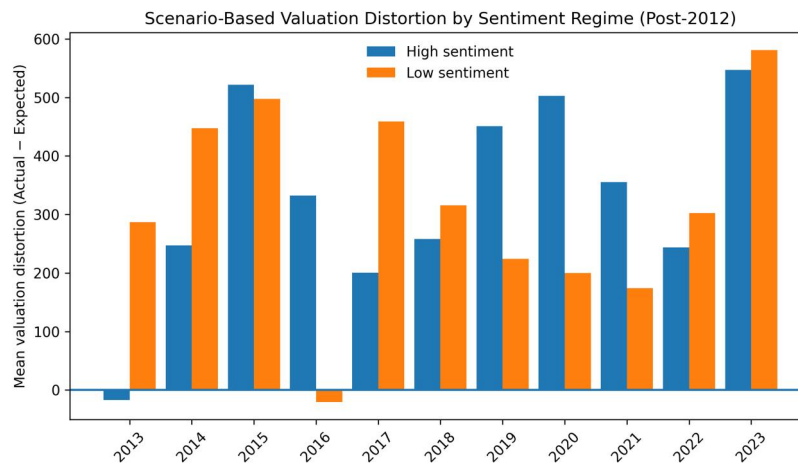


Figure 8. Scenario-based valuation distortion under high vs low sentiment regimes.

Table 14. Robustness test – returns based specification.

Model	RMSE	MAE	Directional accuracy
Random Forest	1.084	0.827	0.422
Linear Regression	1.073	0.811	0.567
Random Walk	1.075	0.818	0.000

4.7. Robustness: returns-based specification

To assess whether the results are sensitive to the choice of dependent variable, we re-estimate the model using log returns instead of price levels.

Table 14 presents the results that indicate weaker predictive performance across all models, reflecting the well-known difficulty of forecasting short-horizon equity returns. In this specification, the linear model slightly outperforms the Random Forest in terms of directional accuracy, while both models exhibit similar error magnitudes. The random walk benchmark continues to show no directional predictive power. This also suggests that the observed performance is not driven by simple trend persistence in price levels, but reflects information contained in the model inputs.

These findings highlight an important distinction in the modelling framework. Price-level modelling allows a direct mapping to valuation frameworks, whereas returns do not provide a natural benchmark for mispricing. The objective of this study is not to predict short-term returns, but to identify valuation distortions through deviations from a model-implied price benchmark. Price-level modelling is therefore more appropriate in this context, as it allows the construction of a conditional expected price against which observed prices can be compared. In contrast, return-based models focus on incremental price changes, which are inherently noisy and less suited for identifying relative mispricing.

Taken together, the results indicate that the main findings are not driven by mechanical trend-following in price levels. Instead, the price-based specification provides a more stable and economically meaningful framework for capturing valuation distortions, while the return-based results serve as a robustness check confirming that the signal does not translate into strong short-horizon return predictability.

5. Discussion and implications

The results of this study provide robust evidence that non-linear, ensemble-based machine learning models, specifically Random Forest (RF), enhance the modelling of expected price dynamics and valuation distortion in Indian equities. Traditional asset pricing frameworks such as discounted cash flow (DCF), residual income models, and the capital asset pricing model (CAPM) remain grounded in assumptions of linearity, rational expectations, and informationally efficient markets (Damodaran, 2012; Stracca, 2004). These assumptions rarely hold in emerging markets, where structural frictions, behavioural anomalies, and episodic institutional flows strongly shape price dynamics. By contrast, the RF framework developed here explicitly incorporates such complexities, demonstrating predictive performance and interpretability in contexts where traditional approaches underperform.

A first major insight is the confirmation that valuation distortion in Indian equities cannot be adequately explained by fundamentals alone. Feature importance analysis consistently ranked sentiment measures (Reddit-derived sentiment and Google Trends indices), institutional flow variables, and promoter holdings alongside fundamentals such as EPS and ROE as leading drivers of expected price estimates (see [Table 13](#); [Figure 7](#)). This echoes findings from behavioural finance literature that emphasise herding, sentiment-driven dynamics, and psychological biases as persistent forces in asset markets (Barberis et al., 1998; Shiller, 2017; Yao et al., 2014). Importantly, the relative weight of behavioural attention varies across market regimes, being more pronounced in bull markets than in downturns, which aligns with the Adaptive Market Hypothesis.

Second, Random Forests demonstrated stable out-of-sample performance under rolling-window evaluation. Unlike linear regression, which often struggles when relationships shift across regimes, the RF model captured non-linear interactions between fundamentals, flows, and behavioural signals. This reinforces arguments in financial machine learning scholarship that ensemble methods provide robustness under regime shifts and distributional drift (Martín-Cervantes & Valls Martínez, 2023; Rezaei et al., 2025). By aggregating across multiple decision trees, RF reduced variance while enabling feature-level interpretability via SHAP values, balancing predictive accuracy with transparency. That said, interpretability remains partial: SHAP values provide decomposition of feature contributions, but they do not establish causal mechanisms in the way econometric models attempt to. This is why ML outputs are best interpreted alongside behavioural finance theory rather than treated as stand-alone causal evidence. In addition, the benchmark comparison against linear regression and a random walk specification confirms that the Random Forest framework provides incremental predictive value, particularly in reducing error magnitudes. This supports the role of non-linear modelling in capturing complex interactions that are not well represented in linear specifications.

Third, the results have market-specific implications. Valuation distortions were especially pronounced in small-cap equities with weaker fundamentals but higher behavioural attention. This reflects the increasing retailisation of Indian equity markets, where low-cost trading platforms and online narratives have amplified the influence of retail investors. These findings contribute to the literature on emerging market anomalies by showing how structural features – such as promoter concentration and episodic FII/DII flows – interact with behavioural signals to shape pricing outcomes. The role of promoter holdings is particularly noteworthy, as it represents a structural factor distinctive to Indian markets. Its explanatory power, alongside liquidity and attention variables, illustrates how market structure itself becomes a driver of valuation distortions.

Fourth, the interpretability analysis through SHAP values and scenario testing provides insight into mechanisms consistent with behavioural theories. Periods of high attention with stable fundamentals are associated with short-term overpricing relative to the expected price benchmark, whereas muted sentiment combined with stronger fundamentals is associated with underpricing. These scenario-based analyses ([Figure 8](#)) illustrate how behavioural dynamics amplify or dampen valuation distortion across regimes. Here, SHAP values function as diagnostic tools that guide theory-building, pointing researchers back to behavioural finance concepts such as overreaction, underreaction, and adaptive heuristics.

Finally, the results carry broader theoretical meaning. They remind us that markets are not the perfectly rational machines described by the Efficient Market Hypothesis (Fama, 1970) (Fama, 1970), nor are they entirely chaotic or irrational. Instead, they reflect a shifting balance shaped by how investors learn, adapt, and sometimes misjudge the environment around them (Lo, 2004; Shiller, 2017; Urquhart & McGroarty, 2016). That behavioural and structural factors explain valuation distortion alongside fundamentals highlights why EMH alone cannot account for markets like India. At the same time, the regime-dependent patterns observed fit closely with the Adaptive Market Hypothesis, which argues that efficiency is not fixed but evolves with context. Importantly, by combining Random Forest models with interpretability tools such as SHAP values, this study shows that machine learning can do more than predict; it can help unpack how theories of efficiency and behaviour play out in real markets. It is also important to note that the robustness analysis using return-based specifications shows weaker predictive performance, reinforcing that the primary contribution of the framework lies in identifying valuation distortions rather than forecasting short-horizon returns. This distinction supports

the interpretation of the model as a conditional expected-price benchmark rather than a trading signal.

In summary, the discussion highlights five core contributions: (i) demonstrating the explanatory weight of behavioural signals in valuation distortion; (ii) establishing the stability of non-linear ensemble models under rolling-window evaluation; (iii) revealing the structural uniqueness of Indian markets via promoter holdings and participation measures; (iv) using SHAP-based interpretability to connect features with valuation distortion patterns while acknowledging limitations; and (v) advancing behavioural finance perspectives by integrating them with machine learning. Together, these contributions position the study within a literature that seeks to decode valuation distortions as an interplay of fundamentals, behaviour, and structure.

6. Policy Implications and future research

6.1. Policy implications

The findings also hold important implications for different stakeholders in India's financial system. For regulators, particularly the Securities and Exchange Board of India (SEBI), the integration of behavioural signals such as search intensity and sentiment indices into surveillance systems could provide early warnings of speculative bubbles and systemic risks. Similar initiatives are already under discussion in Europe and the United States, where regulators are experimenting with machine learning tools to monitor investor sentiment and market fragility. Embedding comparable approaches in India would strengthen market oversight and align local practices with global regulatory trends (Charith & Azeez, 2025; Yao et al., 2014). Such interventions also resonate with the broader ethical-finance perspective, which argues that valuation models should account for social and utility dimensions rather than focusing exclusively on financial efficiency (Stiefenhofer, 2021). Institutional investors can also benefit by supplementing their valuation frameworks with behavioural and structural variables. Promoter holdings and retail sentiment, in particular, have been shown to provide leading signals of valuation distortion in small-cap segments, which are often neglected by conventional asset-pricing models. Incorporating such indicators into risk management systems could help institutional portfolios avoid exposure to inflated stocks and take advantage of underpricing when fundamentals are strong. Recent work has shown how machine learning-based risk management tools can systematically integrate such signals, offering investors a more adaptive and data-rich approach (Ti et al., 2024). At the retail level, the study underlines the importance of investor education. Sentiment-driven surges in trading interest – captured through Google Trends or online forum chatter – tend to amplify valuation distortions, exposing small investors to higher risk. Awareness campaigns coordinated by SEBI or stock exchanges could highlight the dangers of momentum-driven speculation and promote longer-term, fundamentals-based investment behaviour. Beyond protecting individual investors, such campaigns would also reduce the likelihood of market-wide bubbles. Overall, the policy lessons are twofold: they give market participants practical tools and help regulators with evidence-based ways to support financial stability.

6.2. Limitations

Despite the contributions, several limitations should be acknowledged. First, the behavioural proxies are restricted to Reddit sentiment and Google Trends data, which, while informative, do not capture the full spectrum of retail behaviour. Future extensions could incorporate real-time news sentiment, intraday liquidity shocks, and broader social media platforms to enrich behavioural coverage. Second, the study is restricted to Indian equities, meaning that findings may not automatically generalise to other emerging or developed markets, where institutional structures and retail dynamics differ.

6.3. Future research directions

Future research could extend this study in several directions. First, expanding behavioural measures to include corporate filings, real-time news sentiment, and alternative data such as mobile payments or

ESG disclosures would improve the robustness of valuation distortion detection. Second, comparative cross-market studies could test whether the behavioural–structural–fundamental nexus observed in India is universal or market-specific, thereby strengthening external validity. Third, developing dynamic monitoring dashboards that integrate streaming data into the RF framework could shift analysis from retrospective evaluation to real-time surveillance. Such tools would support regulators like SEBI in identifying systemic risks, assist institutional investors in spotting trading opportunities, and help retail participants make more informed decisions. In sum, these directions highlight the dual contribution of this study: offering actionable insights for market participants while setting a forward-looking research agenda at the intersection of behavioural finance and machine learning.

7. Conclusion

This study set out to explain valuation distortion in Indian equities by combining fundamentals, structural features, market data, and behavioural signals within a Random Forest expected-price framework. Unlike traditional valuation methods such as DCF, residual income models, or CAPM – which rest on assumptions of linearity and market efficiency (Damodaran, 2012; Stracca, 2004). The RF model captures non-linear dynamics, adaptive behaviour, and structural frictions inherent in emerging markets.

The findings confirm that the framework achieves stable out-of-sample predictive performance under rolling-window evaluation, with strong Directional Accuracy and consistent error metrics across time windows (Tables 9 and 10). Valuation distortion patterns were especially pronounced in small-cap stocks with weaker fundamentals yet elevated behavioural attention, illustrating the influence of retail-driven dynamics in India’s increasingly digitalised equity markets. Feature importance analysis highlighted EPS, trading activity, institutional flows, promoter holdings, and behavioural attention variables as dominant predictors, consistent with behavioural finance theories such as herding and sentiment bias (Barberis et al., 1998; Shiller, 2017).

The study’s contributions are threefold. First, it demonstrates how integrating fundamentals, structural features, and behavioural indicators can improve expected price estimation in an emerging market setting. Second, it provides methodological value by showing how Random Forests, coupled with SHAP-based interpretability, balance predictive accuracy with explanatory insight. Third, it advances theory by operationalising the Adaptive Market Hypothesis, illustrating how behavioural signals interact with fundamentals across regimes (Lo, 2004; Urquhart & McGroarty, 2016). Importantly, the inclusion of a return-based payoff validation provides an additional economic check on the valuation distortion signal, complementing purely statistical accuracy measures.

From a practical perspective, the findings offer regulators, institutional investors, and retail participants actionable insights. Regulators can adopt behavioural signals into monitoring frameworks to detect excessive overpricing and underpricing early. Institutional investors can enhance valuation models with non-traditional indicators to improve portfolio decisions. Retail investors can better understand the risks of sentiment-driven trading and avoid speculative excesses.

As discussed, the study faces limitations related to behavioural proxy coverage, market-specificity, and interpretability, while future research can explore new data sources, richer behavioural measures, and cross-market replication. Nonetheless, the evidence supports the view that machine learning, when integrated with behavioural and structural finance, provides a useful framework for understanding valuation distortions in emerging equity markets. In conclusion, this study proposes a behavioural–machine learning valuation system that enriches both academic research and practical investment. By bridging predictive modelling with theory-driven interpretation, it offers a pathway for more adaptive and actionable approaches to asset pricing in the twenty-first century.

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All the authors contributed to the manuscript in different capacities.

Author contributions

CRedit: **Vinodh K. Natarajan**: Conceptualization, Methodology, Writing – original draft; **Mahmood Ali Akbar**: Writing – review & editing; **Dalal Aloumi**: Formal analysis, Investigation, Validation; **Ahmed A. Khudhur**: Data curation, Formal analysis.

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Data availability statement

All data used in this study were sourced exclusively from publicly accessible databases, peer-reviewed literature, and institutional reports. This approach ensures full reproducibility and reflects a deliberate commitment to open-source research practices that support transparency, equity, and scholarly rigor. Requests to access the datasets should be directed to the authors.

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